- Compute $\cos(w_k(a+b))$ and $\sin(w_k(a+b))$ using the trigonometric identities:
$$\cos(w_k(a+b)) = \cos(w_k a)\cos(w_k a) - \sin(w_k a)\sin(w_k b)$$
$$\sin(w_k(a+b)) = \sin(w_k a)\cos(w_k b) + \cos(w_k a)\sin(w_k b)$$
In our networks, this is computed in the attention and MLP layers.
- For each output logit c, compute $\cos(w_k(a+b-c))$ using the trigonometric identity:
$$\cos(w_k(a+b-c)) = \cos(w_k(a+b))\cos(w_k c) + \sin(w_k(a+b))\sin(w_k c). \quad (1)$$
This is a linear function of the already-computed values $\cos(w_k(a+b))$, $\sin(w_k(a+b))$ and is implemented in the product of the output and unembedding matrices W_L.
- The unembedding matrix also adds together $\cos(w_k(a+b-c))$ for the various ks. This causes the cosine waves to constructively interfere at $c^* = a + b \mod p$ (giving c^* a large logit), and destructively interfere everywhere else (thus giving small logits to other cs).

We refer to this algorithm as Fourier multiplication, and will justify our claim in detail in Section 4.

4 Reverse engineering a one-layer transformer

In this section, we describe four lines of evidence that our transformers are using the Fourier multiplication algorithm described in Section 3.1. Here we apply our analysis to the mainline model from Section 3; the results are broadly consistent for other models, including across different number of layers, different fractions of the training data, and different prime moduli (see Appendix C.2, especially Table 5).

Our first line of evidence involves examining the network weights and activations and observing consistent **periodic structure** that is unlikely to occur by chance (Section 4.1). Moreover, when we take Fourier transforms, many components are either sparse or nearly sparse in the Fourier domain, supported on a handful of *key frequencies*.

We next look into the actual **mechanisms** implemented in the model weights (Section 4.2). We show that the unembedding matrix W_L is (approximately) rank 10, where each direction corresponds to the cosine or sine of one of 5 key frequencies. Projecting the MLP activations onto the components of W_L approximately produces multiples of the functions $\cos(w_k(a+b))$ and $\sin(w_k(a+b))$, showing that the MLP layer does compute these sums.

To better understand the mechanism, we **zoom in** to individual neurons (Section 4.3). We find that the attention heads and most neurons are well-approximated by degree-2 polynomials of sines and cosines at a *single* frequency. Moreover, the corresponding direction in W_L also contains only that frequency. This suggests that the model's computations are (1) localized across frequencies and (2) mostly aligned with the neuron basis.

Finally, we use **ablations** to confirm that our interpretation is faithful (Section 4.4). We replace various components of the model by the components of the Fourier multiplication algorithm and find that doing so consistently does not harm and sometimes even improves model performance.

4.1 Suggestive evidence: surprising periodicity

The first line of evidence that the network is using the algorithm described in Section 3.1 is the surprising periodicity in the activations of the transformer. That is, the output of every part of the network is periodic as a function of the input tokens.

Periodicity in the embeddings. We start by examining the embeddings. We apply a Fourier transform along the input dimension of the embedding matrix W_E then compute the ℓ_2-norm along the other dimension; results are shown in Figure 3. We plot only the components for the first 56 frequencies, as the norm of the components for frequencies k and $P-k$ are symmetric. The embedding matrix W_E is sparse in the Fourier basis–it only has significant nonnegligible norm at 6 frequencies. Of these frequencies, only 5 appear to be used significantly in later parts of the model (corresponding to $k \in \{14, 35, 41, 42, 52\}$). We dub these the *key frequencies* of the model.

Periodicity in attention heads and MLP neuron activations. This periodic structure recurs throughout the network. As an example, we plot the attention weight at position 0 for every combination of two inputs for head 0 in Figure 4. The attention exhibits a periodic structure with frequency

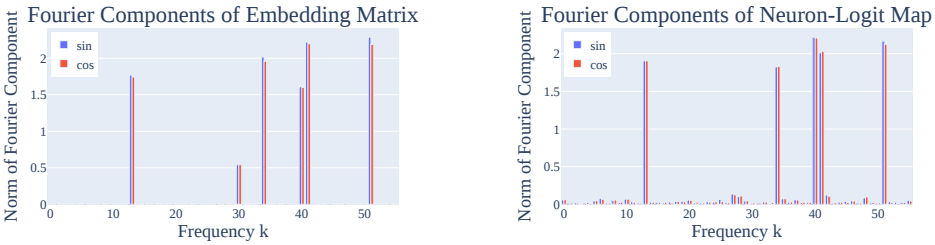

Figure 3: *(Left)* The norms of the Fourier components in the embedding matrix W_E. As discussed in Section 4.1, the sparsity of W_E in the Fourier basis is evidence that the network is operating in this basis. Of the six non-zero frequencies, five "key frequencies" appear in later parts of the network, corresponding to $k \in \{14, 35, 41, 42, 52\}$. *(Right)* Norm of Fourier components of the neuron-logit map W_L. A Fourier transform is taken over the logit axis, and then the norm is taken over the neuron axis. As discussed in Section 4.2, W_L is well-approximated by the 5 key frequencies w_k.

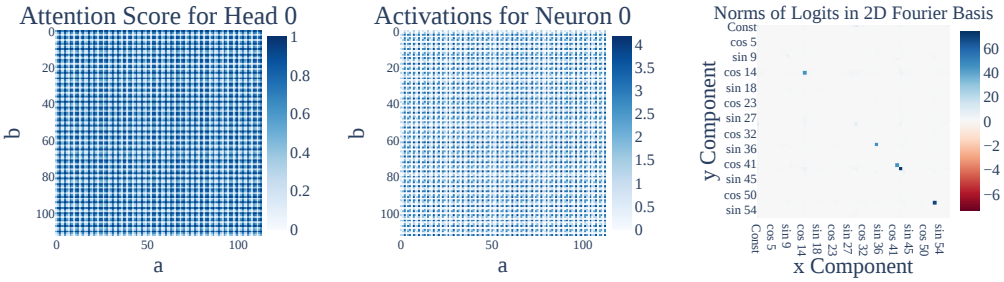

Figure 4: *(Left)* The attention score for head 0 from the token '=' to 'a', as a function of inputs a, b. *(Center)* The activations of MLP neuron 0 given inputs a, b. Both the attention scores and the neuron activations are periodic (Section 4.1). *(Right)* The norm of the Fourier components of the logits (2D Fourier transform is taken over the inputs a, b, and then norm is taken over the logit axis). There are 20 significant components corresponding to the 5 key frequencies (Section 4.1).

$k = 35$. In Figure 4, we also plot the activations of MLP neuron 0 for every combination of inputs. The activations are periodic with frequency $k = 42$. We see similar patterns for other attention heads and MLP neurons (Appendix C.1).

Periodicity in logits. Finally, the logits are also periodic. In Figure 4, we represent the logits in the 2D Fourier basis over the inputs, then take the ℓ_2-norm over the output dimension. There are only twenty components with significant norm, corresponding to the products of sines and cosines for the five key frequencies w_k. These show up as five 2×2 blocks in Figure 4.

4.2 Mechanistic Evidence: Composing Model Weights

We now demonstrate that the model implements the trigonometric identity (1) as follows: the functions $\cos(w_k(a+b))$, $\sin(w_k(a+b))$ are linearly represented in the MLP activations, and the unembed matrix reads these linear directions and multiplies them by $\cos(w_k c)$, $\sin(w_k c)$ respectively.

We will do this in two steps. First, we show that W_L (the matrix mapping MLP activations to logits) is (approximately) rank 10 and can be well approximated as:

$$W_L = \sum\nolimits_{k \in \{14,35,41,42,52\}} \cos(w_k)\, u_k^T + \sin(w_k)\, v_k^T \tag{2}$$

for some $u_k, v_k \in \mathbb{R}^{512}$, where $\cos(w_k), \sin(w_k) \in \mathbb{R}^{113}$ are vectors whose cth entry is $\cos(w_k c)$ and $\sin(w_k c)$. Second, note that our model implements the logits for a, b as:

$$\text{Logits}(a,b) = W_L \text{MLP}(a,b) \approx \sum\nolimits_k \cos(w_k)\, u_k^T \text{MLP}(a,b) + \sin(w_k)\, v_k^T \text{MLP}(a,b) \tag{3}$$

We check empirically that the terms $u_k^T \text{MLP}(a,b)$ and $v_k^T \text{MLP}(a,b)$ are approximate multiples of $\cos(w_k(a+b))$ and $\sin(w_k(a+b))$ ($> 90\%$ of variance explained). Thus the network computes trigonometric functions in the MLP and reads them off as claimed. As a sanity check, we confirm

W_L Component	Fourier components of u_k^TMLP(a,b) or v_k^TMLP(a,b)	FVE
$\cos(w_{14}c)$	$44.6\cos(w_{14}a)\cos(w_{14}b) - 43.6\sin(w_{14}a)\sin(w_{14}b) \approx 44.1\cos(w_{14}(a+b))$	93.2%
$\sin(w_{14}c)$	$44.1\sin(w_{14}a)\cos(w_{14}b) + 44.1\cos(w_{14}a)\sin(w_{14}b) \approx 44.1\sin(w_{14}(a+b))$	93.5%
$\cos(w_{35}c)$	$40.7\cos(w_{35}a)\cos(w_{35}b) - 43.6\sin(w_{35}a)\sin(w_{35}b) \approx 42.2\cos(w_{35}(a+b))$	96.8%
$\sin(w_{35}c)$	$41.8\sin(w_{35}a)\cos(w_{35}b) + 41.8\cos(w_{35}a)\sin(w_{35}b) \approx 41.8\sin(w_{35}(a+b))$	96.5%
$\cos(w_{41}c)$	$44.8\cos(w_{41}a)\cos(w_{41}b) - 44.8\sin(w_{41}a)\sin(w_{41}b) \approx 44.8\cos(w_{41}(a+b))$	97.0%
$\sin(w_{41}c)$	$44.5\sin(w_{41}a)\cos(w_{41}b) + 44.5\cos(w_{41}a)\sin(w_{41}b) \approx 44.5\sin(w_{41}(a+b))$	97.0%
$\cos(w_{42}c)$	$64.6\cos(w_{42}a)\cos(w_{42}b) - 68.5\sin(w_{42}a)\sin(w_{42}b) \approx 66.6\cos(w_{42}(a+b))$	96.4%
$\sin(w_{42}c)$	$67.8\sin(w_{42}a)\cos(w_{42}b) + 67.8\cos(w_{42}a)\sin(w_{42}b) \approx 67.8\sin(w_{42}(a+b))$	96.4%
$\cos(w_{52}c)$	$60.5\cos(w_{52}a)\cos(w_{52}b) - 65.5\sin(w_{52}a)\sin(w_{52}b) \approx 63.0\cos(w_{52}(a+b))$	97.4%
$\sin(w_{52}c)$	$64.5\sin(w_{52}a)\cos(w_{52}b) + 64.5\cos(w_{52}a)\sin(w_{52}b) \approx 64.5\sin(w_{52}(a+b))$	98.2%

Table 1: For each of the directions u_k or v_k (corresponding to the $\cos(w_k)$ and $\sin(w_k)$ components respectively) in the unembedding matrix, we take the dot product of the MLP activations with that direction, then perform a Fourier transform (middle column; only two largest coefficients shown). We then compute the fraction of variance explained (FVE) if we replace the projection with a single term proportional to $\cos(w_k(a+b))$ or $\sin(w_k(a+b))$, and find that it is consistently close to 1.

that the logits are indeed well-approximated by terms of the form $\cos(w_k(a+b-c))$ (95% of variance explained).

W_L is well approximated by $\cos(w_k c)$ and $\sin(w_k c)$. We perform a discrete Fourier transform (DFT) on the logit axis of W_L and look at the 10 directions u_k, v_k corresponding to $\sin(w_k)$ and $\cos(w_k)$. When we approximate W_L with $\sum_{k\in\{14,35,41,42,52\}} \cos(w_k)\, u_k^T + \sin(w_k)\, v_k^T$, the residual has Frobenius norm that is under 0.55% of the norm of W_L. This shows that W_L is well approximated by the 10 directions corresponding to $\cos(w_k)$ and $\sin(w_k)$ for each of the five key frequencies. We also plot the norms of each direction in Figure 3, and find that no Fourier component outside the 5 key frequencies has significant norm.

The unembedding matrix "reads off" terms of the form $\cos(w_k(a+b))$ and $\sin(w_k(a+b))$ from the MLP neurons. Next, we take the dot product of the MLP activations with each of the directions u_k, v_k for $k \in \{14, 35, 41, 42, 52\}$. Table 1 displays the results: the dot products u_k^TMLP(a,b) and v_k^TMLP(a,b) are well approximated by a multiple of terms of the form

$$\cos(w_k(a+b)) = \cos(w_k a)\cos(w_k b) - \sin(w_k a)\sin(w_k b)\,, \text{ and}$$
$$\sin(w_k(a+b)) = \sin(w_k a)\cos(w_k b) + \cos(w_k a)\sin(w_k b)\,.$$

That is, for each key frequency k, u_k and v_k are linear directions in the space of MLP neuron activations that represent $\cos(w_k(a+b))$ and $\sin(w_k(a+b))$.

Logits are well approximated by a weighted sum of $\cos(w_k(a+b-c))$s. We approximate the output logits as the sum $\sum_k \alpha_k \cos(w_k(a+b-c))$ for $k \in \{14, 35, 41, 42, 52\}$ and fit the coefficients α_k via ordinary least squares. This approximation explains 95% of the variance in the original logits. This is surprising—the output logits are a $113 \cdot 113 \cdot 113$ dimensional vector, but are well-approximated with just the 5 directions predicted by our interpretation. If we evaluate test loss using this logit approximation, we actually see an *improvement* in loss, from $2.4 \cdot 10^{-7}$ to $4.7 \cdot 10^{-8}$.

Taken together, these results confirm that the model computes sums of terms of the form $\cos(w_k(a+b-c)) = \cos(w_k(a+b))\cos(w_k c) + \sin(w_k(a+b))\sin(w_k c)$.

4.3 Zooming In: Approximating Neurons with Sines and Cosines

In the previous section, we showed how the model computes its final logits by using W_L to "read off" trigonometric identities represented in the MLP neurons. We now examine the attention heads and MLP neurons to understand how the identities come to be represented at the MLP layer. In Appendix C.1.2, we show that two of the attention heads approximately compute degree-2 polynomials of sines and cosines of a particular frequency (and the other two are used to increase the magnitude of the input embeddings in the residual stream). Here, we show that most neurons are also well-approximated by degree-2 polynomials, and the map from neurons to logits is localized by frequency.

Most MLP neurons approximately compute a degree-2 polynomial of a single frequency. We next try to approximate the activations of each MLP neuron by a degree-2 polynomial of one of the 5 key frequencies. As shown in Figure 5, out of 512 total neurons, 433 (84.6%) have over 85% of their variance explained with a single frequency.